

ENGINEERING EXPLORATION

ENGINEERING-TURNING IDEAS INTO REALITY

“NECESSITY IS THE MOTHER OF INVENTION”

- Engineering is the practical application of science and mathematics to solve real-world problems. Engineers use scientific knowledge and mathematical principles to design, improve, and innovate technologies that make our lives safer, easier, and more efficient.
- The conveniences we enjoy today were not always available. A few decades ago, everyday tasks required much more time and effort. Today, we cook using modern gas and induction stoves instead of traditional wood-burning stoves. We stay cool with air conditioners instead of relying only on table fans. We communicate instantly through smartphones instead of depending on landline telephones. These advancements are the result of the dedication and creativity of engineers and scientists.

- Engineers carefully observe the challenges people face in their daily lives. They analyse these problems, improve existing solutions, and develop new technologies that enhance our quality of life. Every bridge we cross, every vehicle we travel in, every mobile phone we use, and every modern medical device reflects the impact of engineering.
- As the renowned engineer and inventor Henry Petroski said, "Science is about knowing; engineering is about doing." This reminds us that engineering is not just about understanding ideas—it is about transforming those ideas into practical solutions that benefit society.
- In simple words, engineering is not merely a profession; it is the art of solving problems and improving lives.

DIFFERENCE BETWEEN SCIENCE AND
ENGINEERING

- As the physicist Richard Feynman once said, "What I cannot create, I do not understand." Science helps us understand the world, while engineering helps us create solutions for it.
- Science is the study of the natural world. It focuses on discovering, observing, and explaining the laws and principles that already exist in nature. Scientists seek answers to questions such as Why does gravity exist?, How do living organisms function?, and What causes electricity? Their goal is to expand human knowledge by uncovering the mysteries of the universe.
- Engineering, on the other hand, is the application of scientific knowledge and mathematics to solve real-world problems. Engineers use the discoveries made by scientists to design, build, improve, and innovate technologies that benefit society.
- In simple terms, science explains how the world works, whereas engineering applies that knowledge to make the world work better.

Science discovers.
Engineering delivers.

DIFFERENCE BETWEEN A
SCIENTIST AND AN ENGINEER

"Scientists discover; engineers deliver."

- Although scientists and engineers work towards improving society, their roles are different yet closely connected.
- A scientist seeks to understand the natural world. Scientists observe natural phenomena, conduct experiments, and develop theories and laws that explain how nature works. Their primary goal is to expand human knowledge by discovering and explaining the principles that already exist in the universe.
- An engineer, on the other hand, uses this scientific knowledge along with mathematics and creativity to solve real-world problems. Engineers design, develop, and improve technologies that make our lives safer, easier, and more efficient.
- For example, scientists discovered the principles of electricity and electromagnetism. Engineers applied those discoveries to create electric motors, household appliances, communication systems, and countless other technologies that we use every day.

A good way to remember the difference is:

Scientists ask, "How does nature work?"

Engineers ask, "How can we use this knowledge to improve people's lives?"

In my opinion, scientists and engineers are not competitors—they are partners in innovation.

Scientific discoveries provide the foundation, and engineering transforms those discoveries into practical solutions that benefit society. Together, they shape the modern world.

NEEDS AND WANTS OF A
COMPUTER SCIENCE ENGINEER IN
TODAY'S WORLD

"Innovation begins where problems meet possibilities."

- In today's digital world, Computer Science Engineers are essential because they develop technologies that meet both the needs and wants of society. They create software, secure digital information, and build systems that support healthcare, education, banking, transportation, and communication—these are our everyday needs.
- At the same time, they also develop technologies that improve our lifestyle and convenience, such as social media platforms, entertainment applications, smart devices, virtual assistants, and AI-powered services. These innovations satisfy people's wants and enhance their daily experiences.

To achieve this, a Computer Science Engineer must build a strong foundation in programming, problem-solving, and core computer science concepts while continuously learning emerging technologies like Artificial Intelligence and cybersecurity.

“Computer Science Engineers don't just solve today's problems—they create tomorrow's possibilities.”

VARIOUS DISCIPLINES OF
ENGINEERING

- ◆ COMPUTER SCIENCE AND ENGINEERING
- ◆ CIVIL ENGINEERING
- ◆ MECHANICAL ENGINEERING
- ◆ ELECTRICAL AND ELECTRONIC ENGINEERING
- ◆ ELECTRONICS AND COMMUNICATION ENGINEERING
- ◆ CHEMICAL ENGINEERING
- ◆ BIOMEDICAL ENGINEERING
- ◆ AEROSPACE ENGINEERING

CASE STUDY

BUILDING A SMART HOSPITAL



Introduction :

A modern smart hospital is much more than a building where doctors treat patients. It combines advanced engineering, medical technology, and intelligent software to provide faster diagnosis, efficient patient care, and better hospital management. No single engineering discipline can build a smart hospital on its own. It is the result of collaboration among multiple engineering branches.

1. Civil Engineering – Building a Safe Hospital

Role

Civil engineers design and construct the hospital infrastructure.

Responsibilities:

- ◆ Design hospital buildings.
- ◆ Plan emergency exits and fire safety.
- ◆ Ensure earthquake-resistant structures.
- ◆ Design water supply, drainage, and road access.

Contribution:

They provide a safe and functional environment for patients and healthcare workers.

2. Electrical Engineering – Powering the Hospital

Role:

Electrical engineers ensure uninterrupted electricity.

Responsibilities:

- Design electrical wiring.
- Install backup generators and UPS systems.
- Manage power distribution.
- Improve energy efficiency.

Contribution:

Critical medical equipment continues to operate even during power failures.

3. Electronics and Communication Engineering (ECE)

Role:

ECE engineers develop electronic medical systems and communication networks.

Responsibilities:

- Install patient monitoring devices.
- Develop ECG and heart-rate monitoring systems.
- Set up hospital communication systems.
- Connect medical devices through IoT.

Contribution:

Doctors receive patient information instantly from different departments.

4. Biomedical Engineering

Role:

Biomedical engineers develop and maintain medical equipment.

- ◆ Responsibilities:
- ◆ Design MRI and CT scanners.
- ◆ Develop ventilators.
- ◆ Maintain dialysis machines.
- ◆ Improve medical instruments.

Contribution:

Doctors obtain accurate medical information for diagnosis and treatment.

5. Computer Science Engineering (CSE)

Role:

Computer Science Engineers make the hospital intelligent and connected.

- ◆ Responsibilities:
- ◆ Develop Hospital Management Systems.
- ◆ Store Electronic Health Records (EHR).
- ◆ Build appointment booking applications.
- ◆ Develop AI to detect diseases from X-rays and scans.
- ◆ Secure patient data through cybersecurity.
- ◆ Analyze health data to improve treatment decisions.
- ◆ Integrate wearable devices for remote patient monitoring.

Contribution:

Doctors spend less time on paperwork and more time treating patients, while patients receive faster and more accurate care.

A SIMPLE OUTLINE OF THESE BRANCHES

Civil Engineering:

Constructs the hospital building and infrastructure.

Electrical Engineering:

Provides reliable electrical power.

Electronics & Communication Engineering:

Connects medical devices and communication systems.

Biomedical Engineering:

Designs and maintains medical equipment.

Computer Science Engineering:

Develops software, AI, cybersecurity, and digital healthcare systems.

Why Computer Science Engineering Is Essential?

Computer Science Engineering connects all hospital systems into one intelligent network.

Without CSE:

- Medical records would remain mostly paper-based.
- AI-assisted diagnosis would not be available.
- Online appointments and telemedicine would be difficult.
- Patient data would be harder to secure and analyze.
- Coordination between departments would be slower.

CSE transforms a traditional hospital into a smart hospital by integrating software, artificial intelligence, cloud computing, cybersecurity, and data analytics.

The role of a Computer Science Engineer is not to replace other engineers but to integrate their technologies into intelligent systems that improve efficiency, accuracy, and the quality of human life.

MISCONCEPTIONS ABOUT
COMPUTER SCIENCE
ENGINEER

"Computer Science Engineers only write code."

Reality:

Coding is just one part of the profession. CS engineers also design systems, solve real-world problems, develop AI, secure digital systems, analyze data, build cloud platforms, create mobile apps, and innovate new technologies

"CS engineers work only for high salaries."

Reality:

While the field offers good career opportunities, many CS engineers are driven by innovation, problem-solving, and the desire to create technologies that improve lives.

"Computer Science Engineers work alone all the time."

Reality:

They collaborate with mechanical, civil, electrical, biomedical, business, and design teams to develop products used by millions of people.

"Computer Science is only for people who are experts in programming."

Reality:

Programming is a skill that is learned over time. Success in computer science also requires logical thinking, creativity, communication, teamwork, and continuous learning.

"Once a software is built, the job is over."

Reality:

CS engineers continuously maintain, improve, secure, and update software to meet changing user needs and protect against new security threats.

"A CS Engineer builds not just software, but the future."

SKILLS REQUIRED FOR CS
ENGINEER IN 21ST CENTURY

- ◆ Problem-Solving – Find smart and efficient solutions.
- ◆ Creative Thinking – Innovate and build new technologies.
- ◆ Collaboration – Work effectively with engineers from other disciplines.
- ◆ Continuous Learning – Adapt to rapidly changing technologies.
- ◆ AI & Digital Literacy – Understand and use AI and modern digital tools.
- ◆ Cybersecurity Awareness – Protect systems and user data.
- ◆ Communication – Explain ideas clearly and work in teams.
- ◆ Ethical Responsibility – Develop technology that benefits society.

"In the 21st century, the greatest strength of a Computer Science Engineer is not mastering one technology—it is the ability to learn, adapt, collaborate, and innovate throughout a lifetime."

THANK YOU.